

The container method

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Theorem 1 (Strong graph container lemma). *For every $c > 0$, there exists $\delta > 0$ such that the following hold. Let G be an n -vertex graph of average degree d and maximum degree $\Delta(G) \leq cd$, and let $\tau = 2\delta/d$. There exists a collection $\mathcal{C}$ of subsets of vertices of G and a function $f : 2^{V(G)} \rightarrow \mathcal{C}$ such that:*

- (i) *every independent set I of G has a fingerprint S such that $S \subseteq I \subseteq f(S)$ and $|S| \leq \tau n$,*
- (ii) *for every $C \in \mathcal{C}$, $|C| \leq (1 - \delta)n$.*

Theorem 2 (Strong hypergraph container lemma). *For every $c > 0$, there exists $\delta > 0$ such that the following hold. Let $\mathcal{H}$ be a 3-uniform n -vertex hypergraph of average degree d and maximum degree $\Delta_1(\mathcal{H}) \leq cd$ and $\Delta_2(\mathcal{H}) \leq c\sqrt{d}$, and let $\tau = \delta/\sqrt{d}$. There exists a collection $\mathcal{C}$ of subsets of vertices of G and a function $f : 2^{V(G)} \rightarrow \mathcal{C}$ such that:*

- (i) *every independent set I of $\mathcal{H}$ has a fingerprint S such that $S \subseteq I \subseteq f(S)$ and $|S| \leq \tau n$,*
- (ii) *for every $C \in \mathcal{C}$, $|C| \leq (1 - \delta)n$.*

1 Supersaturation of triangles

Exercise 1. Goodman bound: a combinatorial proof of supersaturation

1. Show that for every graph G , $\sum_{uv \in E(G)} \deg(u) + \deg(v) = \sum_{u \in V(G)} \deg(u)^2$
2. For every u and v , denote $\deg(u, v)$ the codegree of u, v , that is the number of common neighbours of u and v . Show that $\sum_{u \in V(G)} \deg(u)^2 \leq |E(G)| |V(G)| + \sum_{uv \in E(G)} \deg(uv)$
3. Show that if G has $\gamma \binom{n}{2}$ edges, then G contains at least $\gamma(2\gamma - 1) \frac{n^3}{6}$ triangles.

Solution 1.

1. Both sides of the equation count the number of triples (a, b, c) of vertices, with ab and $bc \in E(G)$.
2. For every edge $uv \in E(G)$, using inclusion-exclusion formula,

$$\begin{aligned} \deg(u) + \deg(v) &= |N(u) \setminus N(v)| + |N(v) \setminus N(u)| + 2|N(u) \cap N(v)| \\ &\leq |V(G)| + \deg(u, v) \end{aligned}$$

Summing this over all edges and using the previous double-counting inequality gives

$$\sum_{u \in V(G)} \deg(u)^2 \leq |E(G)| |V(G)| + \sum_{uv \in E(G)} \deg(uv)$$

3. By convexity, $\frac{1}{n} \sum_{u \in V(G)} \deg(u)^2 \geq \left(\frac{\sum_{u \in V(G)} \deg(u)}{n} \right)^2 = \left(\frac{2|E(G)|}{|V(G)|} \right)^2$. Note that for every edge uv , the codegree $\deg(uv)$ counts the number of triangles that contain u and v . Hence, the total number of

triangles in G is

$$\begin{aligned}
T(G) &= \frac{1}{3} \sum_{uv \in E(G)} \deg(u, v) \\
&\geq \frac{1}{3} \left(\frac{4|E(G)|^2}{n} - |E(G)|n \right) \\
&\geq \frac{1}{3} \left(\gamma^2 n^3 - \frac{\gamma}{2} n^3 \right) = \gamma(2\gamma - 1) \frac{n^3}{6}
\end{aligned}$$

Exercise 2. A simple probabilistic proof

Let $\varepsilon > 0$ and C be some large constant. Let G be a graph on $n \geq C$ vertices and at least $(\frac{1}{2} + \varepsilon) \binom{n}{2}$ edges.

1. Prove the reversed Markov inequality: let X be a random variable taking values in $[0, 1]$,

$$\mathbb{P}(X \geq t) \geq \frac{\mathbb{E}[X] - t}{1 - t} \text{ for every } t \in (0, 1).$$

2. Let M be a random uniform subset of C vertices. Give a lower bound on the probability that $G[M]$ contains a triangle.
3. Let F be a random uniform triple of distinct vertices of G . Give a lower bound on the probability that $G[F]$ is a triangle.
4. Deduce that the number of triangles in G is $\Omega(n^3)$.

Solution 2.

1. We have $\mathbb{E}[X] \leq (1 - \mathbb{P}(X \geq t))t + \mathbb{P}(X \geq t)$.
2. Let C be a sufficiently large constant such that $\frac{\text{ex}(C, K_3) + 1}{\binom{C}{2}} \leq \frac{1 + \varepsilon}{2}$. Let M be a random uniform set of C vertices of G . Let X be the number of edges in $G[M]$ and $X' = X / \binom{n}{2}$. By linearity of expectation, $\mathbb{E}[X] = \sum_{uv \in E(G)} \frac{\binom{C-2}{n-2}}{\binom{n}{n-2}} = \sum_{uv \in E(G)} \frac{\binom{C}{2}}{\binom{n}{2}} \geq (\frac{1}{2} + \varepsilon) \binom{C}{2}$, so $\mathbb{E}[X'] \geq \frac{1}{2} + \varepsilon$. We have

$$\begin{aligned}
\mathbb{P}(G[M] \text{ contains a triangle}) &\geq \mathbb{P}(X > \text{ex}(C, K_3)) \\
&\geq \mathbb{P}(X' \geq \frac{1 + \varepsilon}{2}) \\
&\geq \frac{\mathbb{E}[X'] - \frac{1 + \varepsilon}{2}}{(1 - \varepsilon)/2} && \text{by Reversed Markov inequality} \\
&\geq \frac{\varepsilon}{1 - \varepsilon} > 0
\end{aligned}$$

3. To sample F , we first sample M and then a random uniform triple of distinct vertices of M . We have

$$\begin{aligned}
\mathbb{P}(F \text{ induces a triangle}) &= \mathbb{P}(F \text{ induces a triangle} | K_3 \subset G[M]) \cdot \mathbb{P}(K_3 \subset G[M]) \\
&\geq \binom{C}{3}^{-1} \frac{\varepsilon}{1 - \varepsilon}
\end{aligned}$$

So the number of triangles in G is

$$T(G) = \mathbb{P}(F \text{ induces a triangle}) \cdot \binom{n}{3} \geq \frac{\varepsilon}{(1 - \varepsilon) \binom{C}{3}} \binom{n}{3}$$

2 Other applications of the container method

Exercise 3.

We admit the following supersaturation result: for every $\varepsilon > 0$, there exists $\gamma > 0$ such that for every subset $A \subset [n]$ of size at least εn , A contains at least γn^2 triples $\{a, b, c\}$ such that $a - b = b - c$.

We denote $[n]_p$ the random subset of $[n]$ containing each element independently with probability p . The goal of this exercise is to prove the following theorem:

Theorem 3. *For every $\varepsilon > 0$, there exists $C > 0$ such that for every $p \geq C/\sqrt{n}$, every subset A of $[n]_p$ with size at least εpn contains an arithmetic progression of length 3.*

1. Prove the following container lemma for arithmetic progressions of length 3: For every γ , there exists $\delta > 0$ such that the following holds for every n . There exists a collection $\mathcal{C}$ of subsets of $[n]$ and a function $f : 2^n \rightarrow \mathcal{C}$ such that:
 - (i) every subset $I \subset [n]$ without arithmetic progression of length 3 has a *fingerprint* S such that $S \subseteq I \subseteq f(S)$ and $|S| \leq 2\delta\sqrt{n}$,
 - (ii) every $C \in \mathcal{C}$ contains at most γn^2 arithmetic progressions of length 3.
2. Prove Theorem 3.

Solution 3.

First, note that the arithmetic progressions of length 3 can be encoded by a 3-uniform hypergraph. Let $\mathcal{H}$ be the hypergraph labeled by $[n]$ and containing an hyperedge for every arithmetic progression of length 3. An independent set of $\mathcal{H}$ is a subset of $[n]$ without arithmetic progression of length 3. The number of edges in $\mathcal{H}$ is $\sum_{k=1}^{\lfloor n/2 \rfloor} n - 2k \sim n^2/2$, so $\mathcal{H}$ has average degree $d \sim n/2$. The maximum degree of $\mathcal{H}$ is $\Delta_1 \leq 3n/2 = 3d$ and we have $\Delta_2 = 3$.

1. Every induced subgraph of $\mathcal{H}$ with at least γn^2 hyperedges has average degree at least γn . Hence one can build a family of fingerprints and container by repeatedly applying the Strong hypergraph container lemma, first to $\mathcal{H}$ with $c = 3$ and $d = n/2$ and then recursively to the containers with at least γn^2 hyperedges with $c = 6/\gamma$ and $d \geq \gamma n$. After t recursive applications, every container has either less than γn^2 hyperedges or at most $(1 - \delta(\gamma))^t n$ vertices. By taking t sufficiently large, the number of hyperedges of each container C , is at most $|C|^2/2 \leq (1 - \delta(\gamma))^{2t} n^2/2 \leq \gamma n^2$.
2. Let $\varepsilon > 0$, $m = \varepsilon pn$ and set $\gamma(\varepsilon/2)$ to be the constant given by the supersaturation theorem applied to $\varepsilon/2$. We apply the container lemma with γ . For every fingerprint S the probability that $[n]_p$ contains a subset $A \supseteq S$ of size εpn and with no arithmetic progression of length 3 is at most $\mathbb{P}(S \subset [n]_p) \cdot \mathbb{P}([n]_p \cap f(S) \setminus S \geq \varepsilon pn - |S|)$. We have $\mathbb{P}(S \subset [n]_p) = p^{|S|}$. As the container $f(S)$ has at most γn^2 arithmetic progressions of length 3, by supersaturation, $|f(S)| \leq \varepsilon n/2$. Hence,

$$\begin{aligned}
 \mathbb{P}(\exists S \subset [n]_p \text{ with no 3-AP and } |S| = m) &\leq \sum_{k=1}^{2\delta\sqrt{n}} \binom{n}{k} p^k \cdot \mathbb{P}(\text{Bin}(\varepsilon n/2 - k, p) \geq m - k) \\
 &\leq \sum_{k=1}^{2\delta\sqrt{n}} \binom{n}{k} p^k e^{-\Omega(\varepsilon np)} && \text{by Chernoff} \\
 &\leq \sum_{k=1}^{2\delta\sqrt{n}} \left(\frac{C\sqrt{n}}{k} \right)^k e^{-\Omega(\sqrt{n})} = o(1)
 \end{aligned}$$

for C enough.

Exercise 4.

The goal of this exercise is to prove the following theorem of Balogh and Solymosi, answering a question of Erdős:

Theorem 4. *There exists a set S of n points in the plane, containing no four points on a line, such that every subset of S of size $n^{5/6+o(1)}$ contains three points on a line.*

To do so, we consider $P = [m]^3$ the lattice in $\mathbb{R}^3$ and admit the following supersaturation theorem for collinear triples: for every $0 < \gamma < 1/2$ and every $S \subset [m]^3$ of size at least $m^{3-\gamma}$, there exist at least $m^{6-4\gamma-o(1)}$ collinear triples of points in S .

1. Prove the following container lemma: For every $m \in \mathbb{N}$, there exists a collection of containers $\mathcal{C}$ of $[m]^3$ such that
 - (i) $|\mathcal{C}| \leq \exp(m^{5/3+o(1)})$
 - (ii) $|C| \leq m^{8/3+o(1)}$ for every $C \in \mathcal{C}$
 - (iii) every set of points S in general position (i.e. without three collinear points) is contained in some $C \in \mathcal{C}$
2. Let $p = m^{-1+o(1)}$ and consider R a random subset of points of $[m]^3$ containing each lattice point independently with probability p . Show that with high probability, R has size $m^{2+o(1)}$, R contains $o(|R|)$ collinear quadruples and for every $C \in \mathcal{C}$, $|R \cap C| \leq m^{5/3+o(1)}$.
3. Prove that there exists $A \subset R$ of size $m^{2+o(1)}$ with no collinear quadruple and containing no subset of $m^{5/3+o(1)}$ points in general position.
4. Finally, argue that A can be projected on a plane such that the projection A and $p(A)$ have the same size and the same number of collinear triples. Conclude.

Solution 4.

1. We encode each collinear triple by an hyperedge in a 3-uniform hyper graph $\mathcal{H}$ on the vertex set $[m]^3$. Let S be an arbitrary subset of $[m]^3$ of size $s \geq m^{8/3+o(1)}$. Let $\gamma = 1/3 + o(1)$ by supersaturation, the number of hyperedges in $\mathcal{H}[S]$ is at least $m^{6-4\gamma-o(1)} = s^4/m^{6+o(1)}$, so $\mathcal{H}[S]$ has average degree at least $d \geq s^3/m^{6+o(1)} = m^{2+o(1)}$. Moreover, the maximum degree $\Delta_2(\mathcal{H}[S])$ is at most the $m \leq \sqrt{d}$.
2. We have $|R| = \text{Bin}(m^3, p)$ so by Chernoff bound, $|R| = m^{2+o(1)}$ with high probability. The number of collinear quadruples in $[m]^3$ is $O(1)$. For every $C \in \mathcal{C}$, $|C| \leq m^{8/3+o(1)}$, so $|C \cap R| \geq m^{(1+\varepsilon)5/3}$ with probability $\exp(-m^{5/3+2\varepsilon+o(1)})$ for every ε by Chernoff bound. So by union bound, $|C \cap R| \leq m^{5/3+o(1)}$ for every $C \in \mathcal{C}$.
3. As there are $o(|R|)$ collinear quadruples in R , one can remove one point in each quadruple, to obtain a set $A \subset R$ of size $m^{2+o(1)}$ with no collinear quadruple and containing no subset of $m^{5/3+o(1)}$ points in general position.
4. We are looking for a projection such that preserves collinear triples of A . Every non-collinear triple (a, b, c) forbids a projection along a vector of the hyperplane containing a, b and c , otherwise, the triple is projected to a collinear triple. Every pair (a, b) forbids a projection along the line passing by a and b , otherwise, a and b are projected to the same point.

★ Exercise 5.

Given a graph G , let $i(G)$ count the number of independent sets of G . The goal of this exercise is prove the following result of Alon: every d -regular graph on n vertices has $i(G) \leq 2^{(1+o(1))n/2}$ independent sets, where $o(1)$ is a function that tends to 0 as d tends to infinity.

1. Let $\beta > 0$. Prove that for every set of vertices A of size at least $\frac{n}{2} + \frac{\beta n^2}{2d}$, we have $e(A) \geq \beta \binom{n}{2}$.
2. Prove that every d -regular graph on n vertices has $i(G) \leq 2^{(1+o(1))n/2}$ independent sets.

★ Solution 5.

1. We have $|A|d = 2e(A) + e(A, \bar{A})$ hence, $e(A) \geq (|A| - |\bar{A}|)d/2 \geq \beta \frac{n^2}{2} \geq \beta \binom{n}{2}$